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(54) **ELECTRICAL CONNECTOR FOR CONNECTING TO A FLAT CABLE AND ELECTRICAL ASSEMBLY USING SAME**

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H01R 12/59 (2011.01)
H01R 12/61 (2011.01)
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CPC H01R 12/78; H01R 12/592; H01R 12/61; H01R 12/774; H01R 13/502; H01R 13/516; H01R 13/639; H01R 12/59; H01R 12/613; H01R 12/77; H01R 12/771; H01R 13/506; H01R 13/62

See application file for complete search history.

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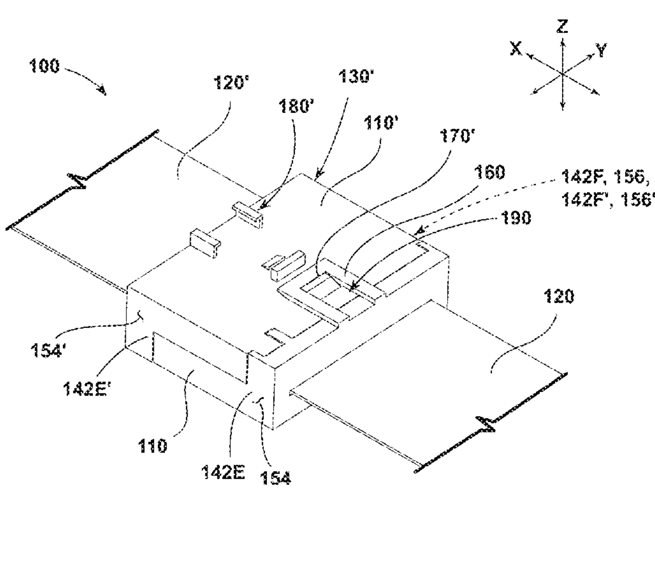
(57) **ABSTRACT**

An electrical connector configured for connection with a flat cable includes a housing and a wedge, in some configurations. The housing may include an internal cavity, a slot in communication with the cavity and configured for receiving said flat cable, and/or a contact plate disposed within the cavity. The wedge may be configured for connection with the contact plate. The housing and the wedge may be configured to secure a portion of said flat cable in the cavity. An electrical assembly may include the electrical connector, a second electrical connector having an identical configuration as the electrical connector, said flat cable, and/or a second flat cable. In an assembled configuration, the second electrical connector may be connected to the electrical connector such that said flat cable is electrically connected to the said second flat cable.

20 Claims, 9 Drawing Sheets



(51) **Int. Cl.**

H01R 13/516	(2006.01)
H01R 13/639	(2006.01)
H01R 13/506	(2006.01)
H01R 13/62	(2006.01)

(52) **U.S. Cl.**
CPC *H01R 12/77* (2013.01); *H01R 12/771* (2013.01); *H01R 13/506* (2013.01); *H01R 13/62* (2013.01)

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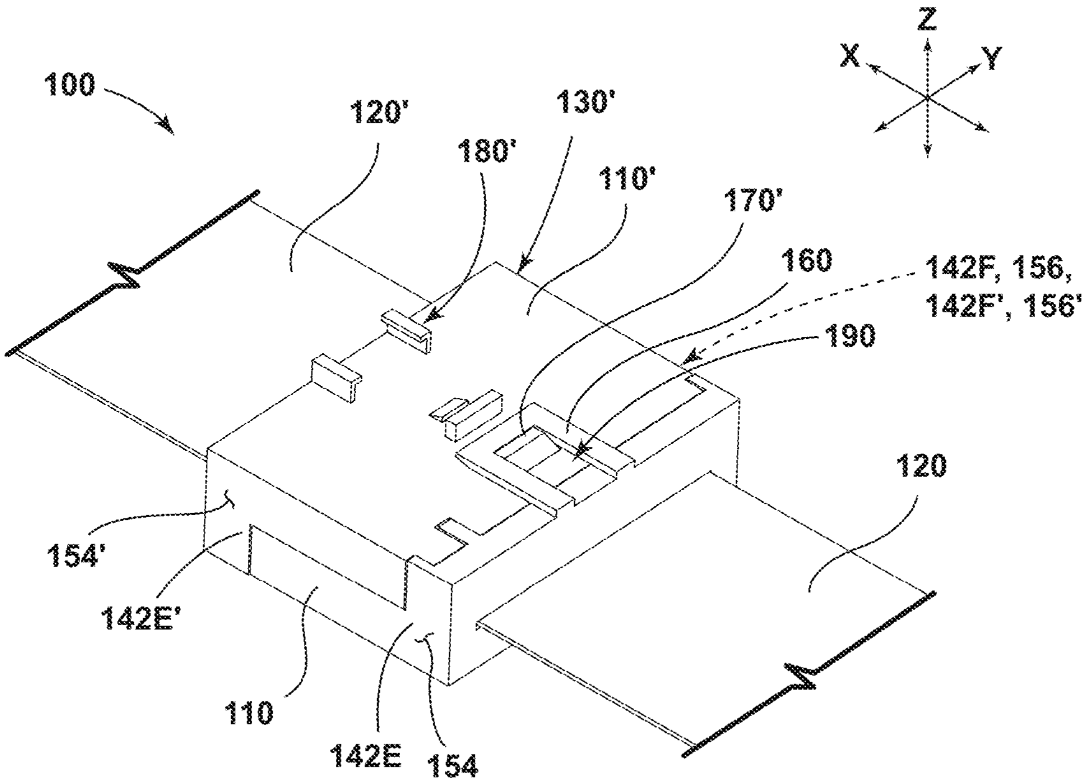
Z
X
Y
100
120'
130'
180'
110'
170'
160
142F, 156,
142F', 156'
190
120
154'
142E'
110
142E
154

FIG. 1

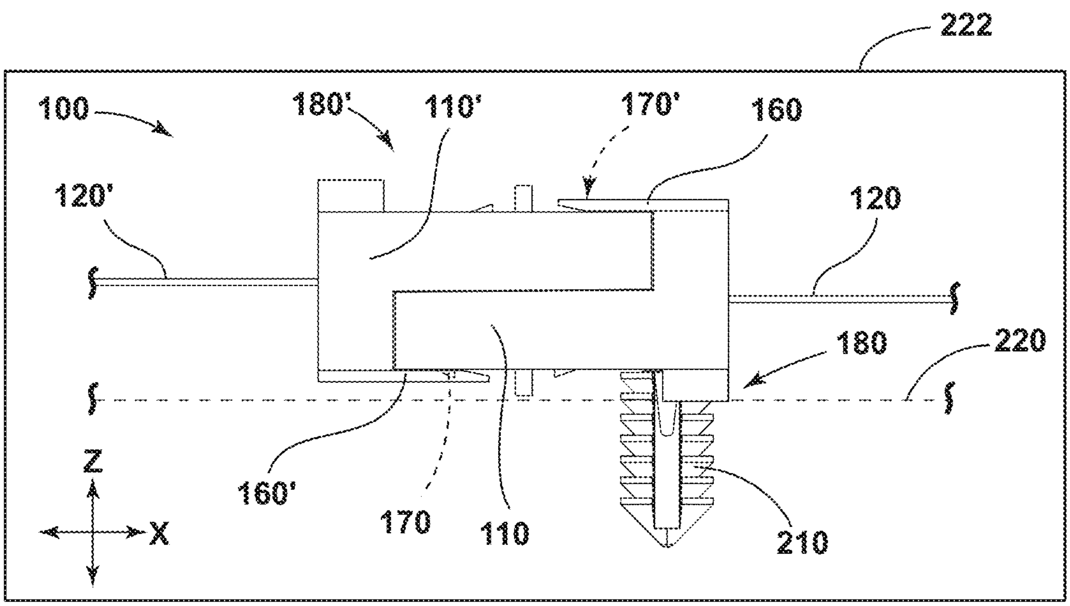
222
100
180'
110'
170'
160
120'
120
180
220
Z
X
160'
170
110
210

FIG. 2

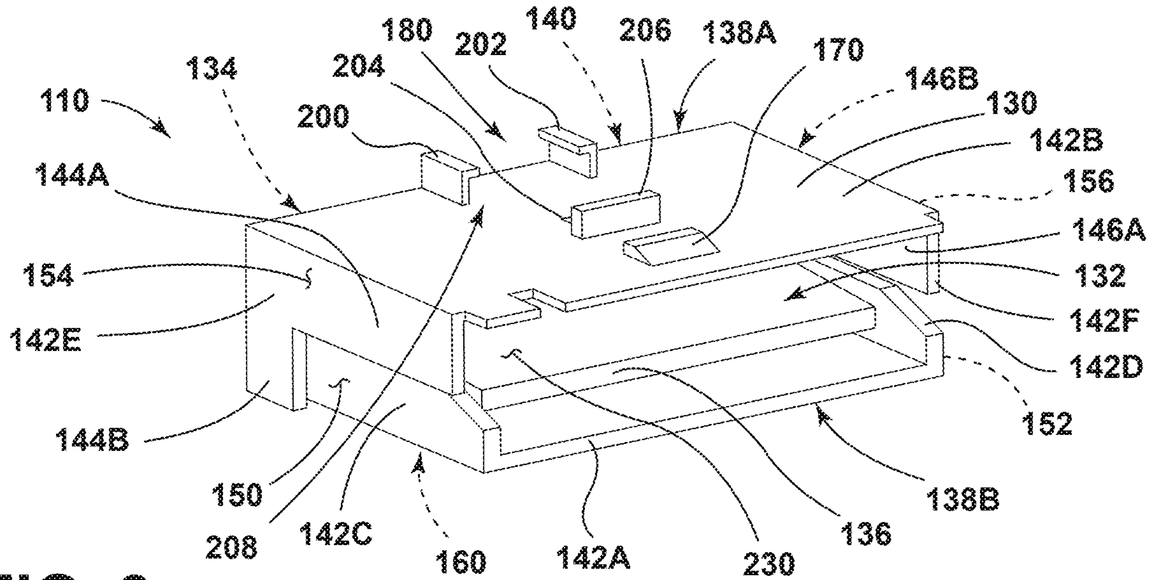

FIG. 3

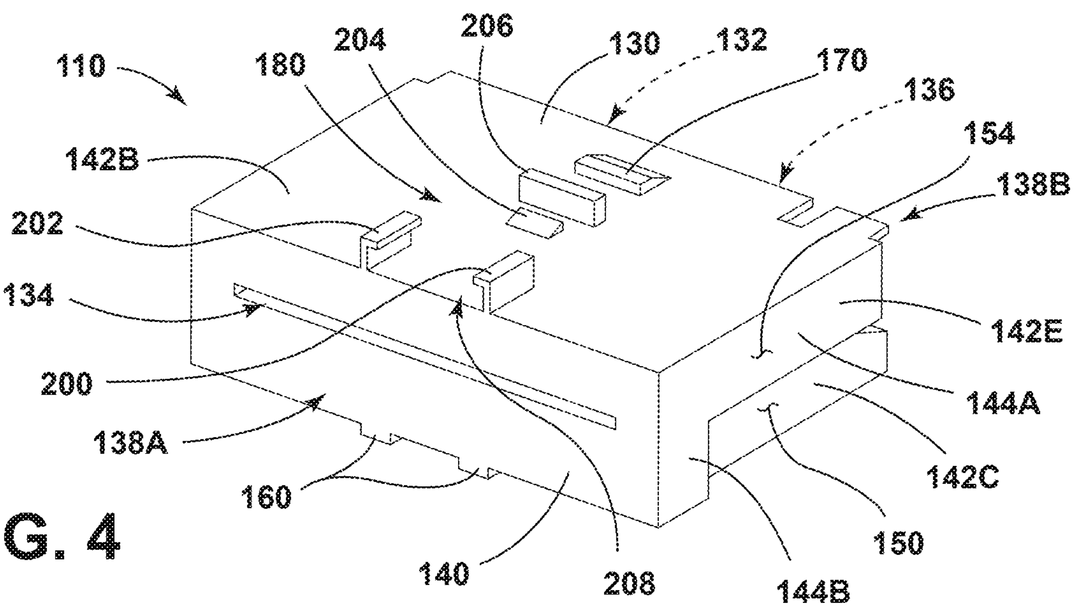

FIG. 4

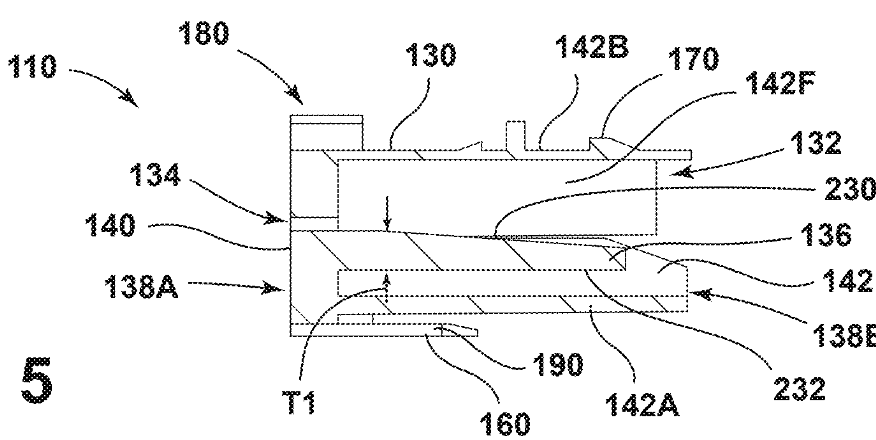

FIG. 5

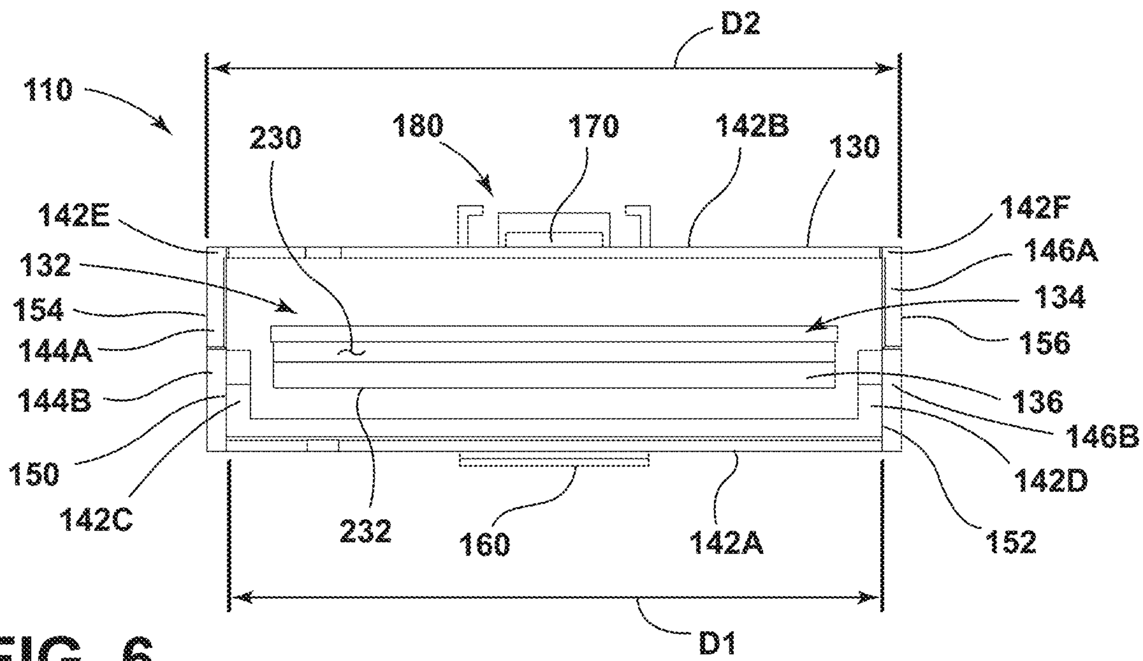
D2
110
230
180
170
142B
130
142E
142F
132
146A
154
134
144A
156
144B
136
146B
150
142D
142C
232
160
142A
152
D1

FIG. 6

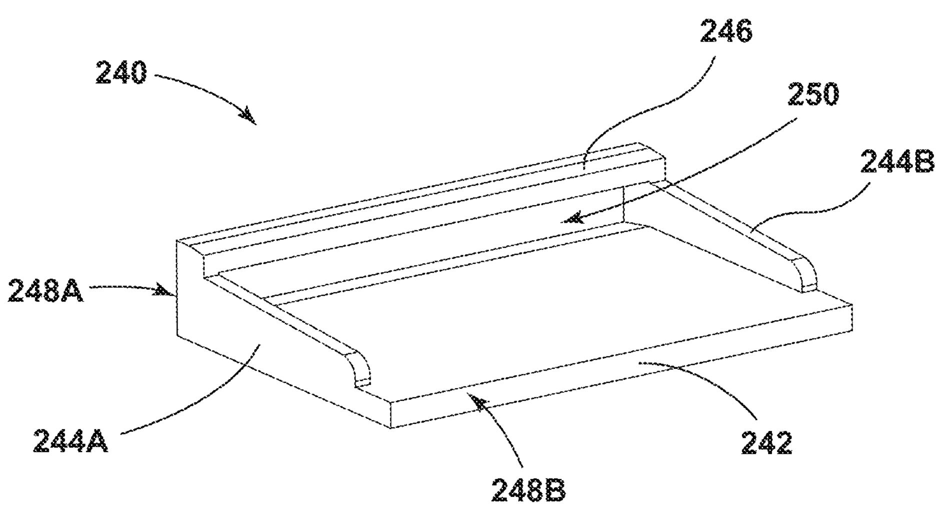
246
240
250
244B
248A
244A
242
248B

FIG. 7

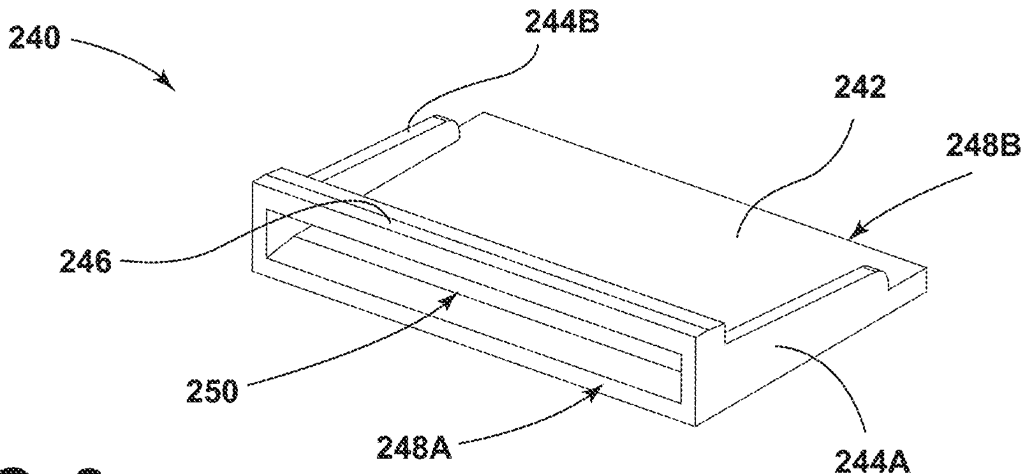
240
244B
242
248B
246
250
248A
244A

FIG. 8

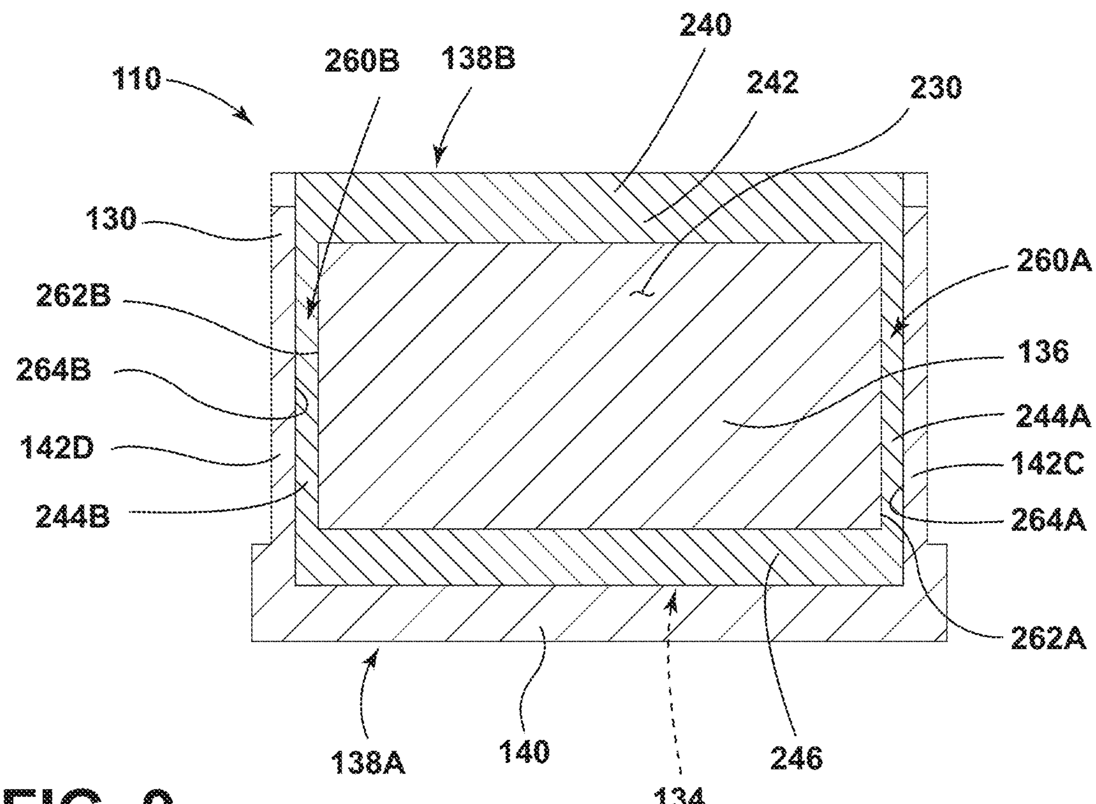
240
110
260B
138B
242
230
130
260A
262B
136
264B
244A
142D
142C
244B
264A
262A
138A
140
246
134

FIG. 9

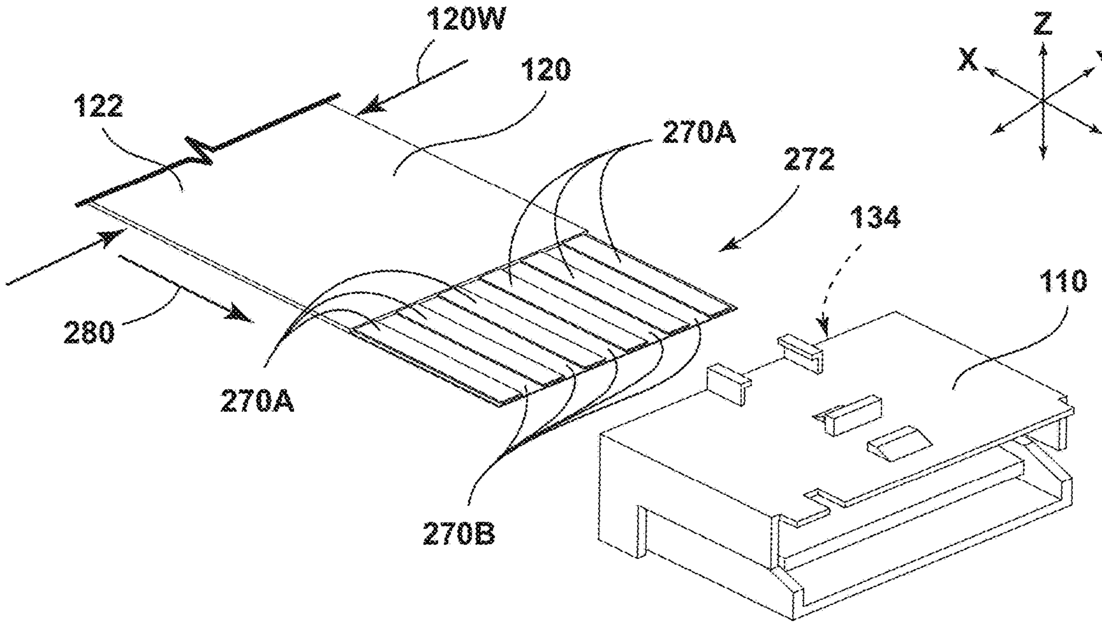
120W
120
122
270A
272
134
110
280
270A
270B
Z
X
Y

FIG. 10

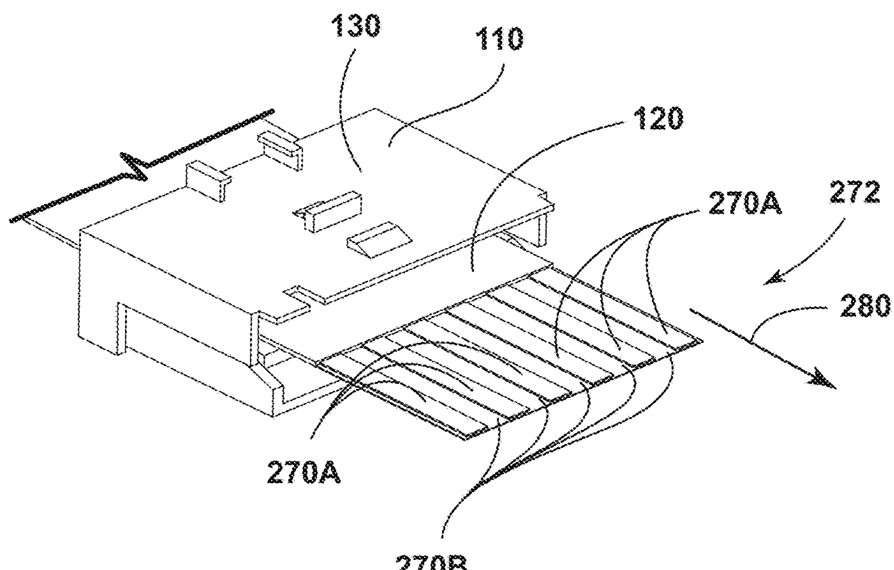
130
110
120
270A
272
280
270A
270B

FIG. 11

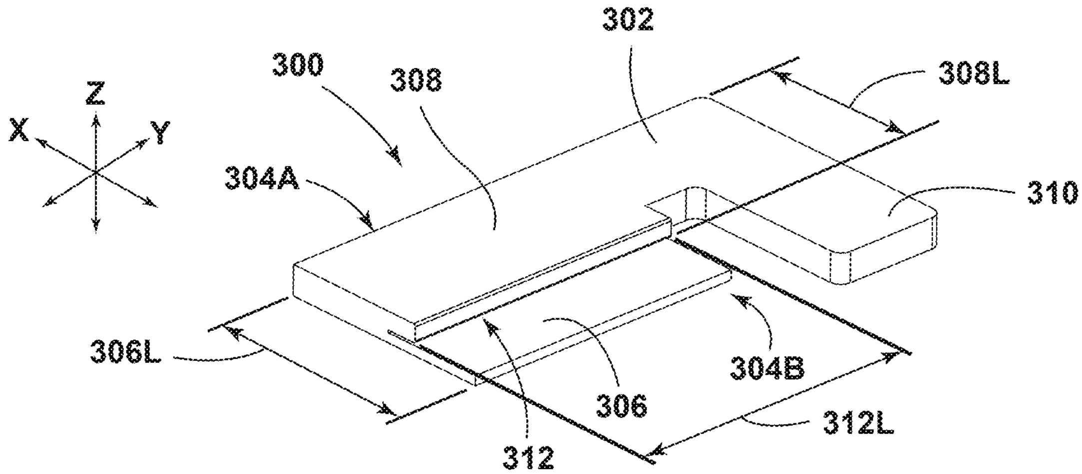

FIG. 12

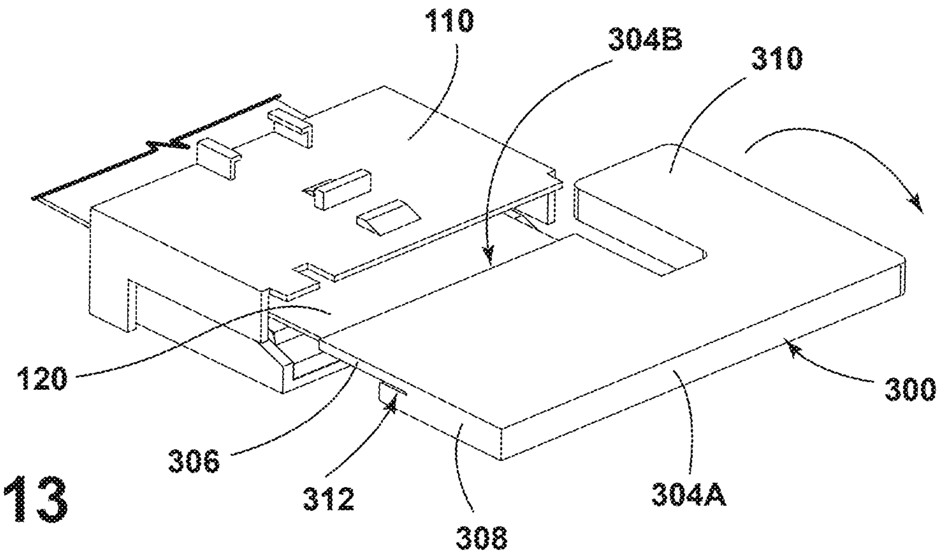

FIG. 13

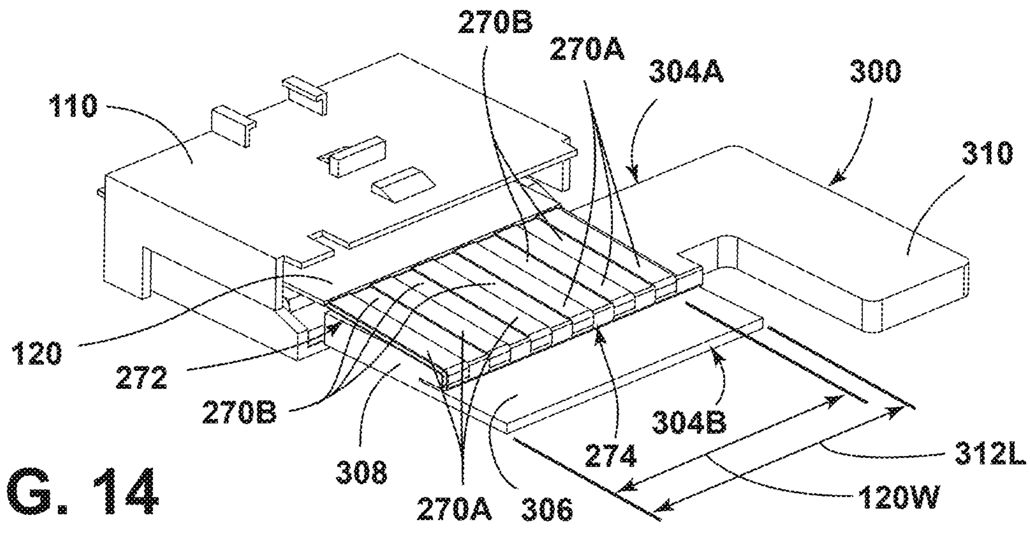

FIG. 14

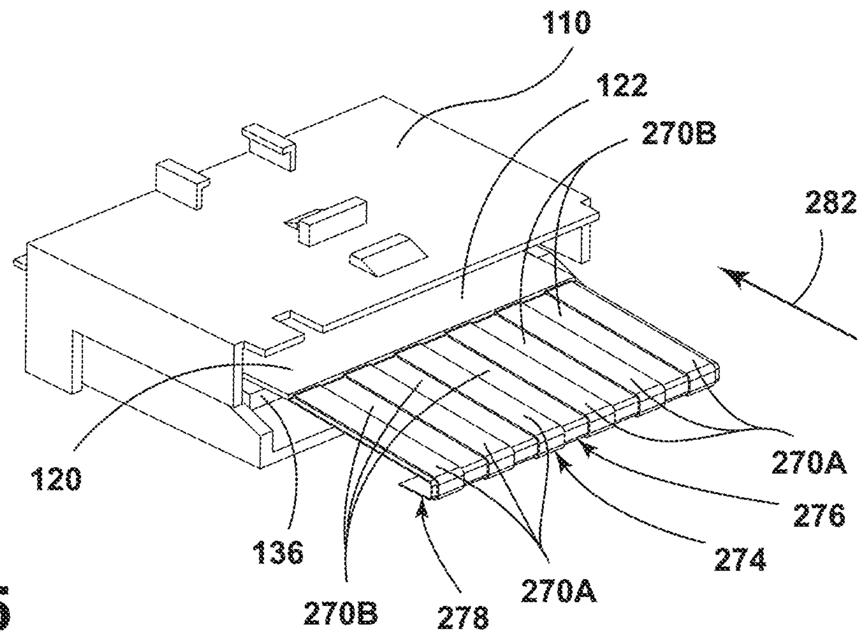
110
122
270B
282
120
136
270B
278
270A
274
276
270A

FIG. 15

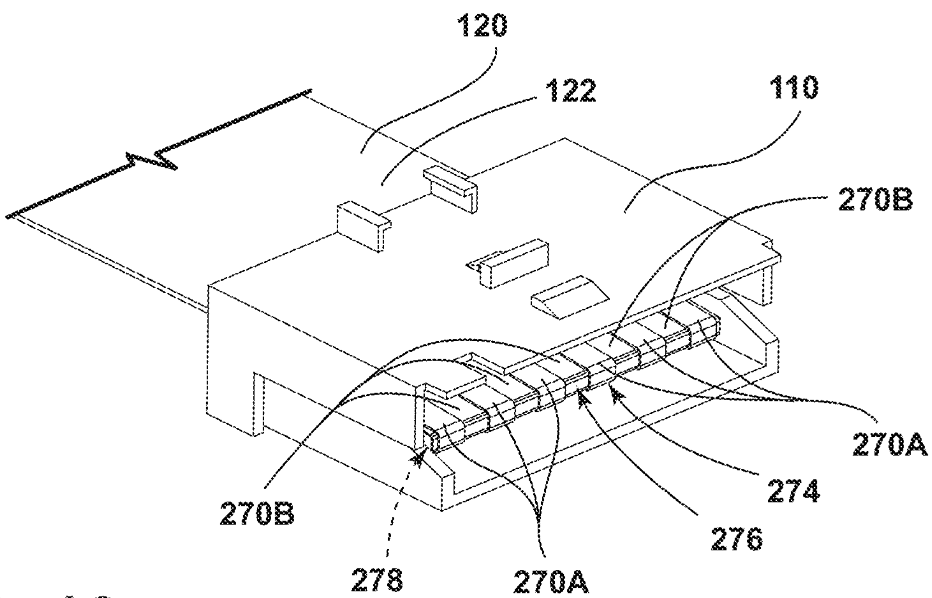
120
122
110
270B
270B
278
270A
276
274
270A

FIG. 16

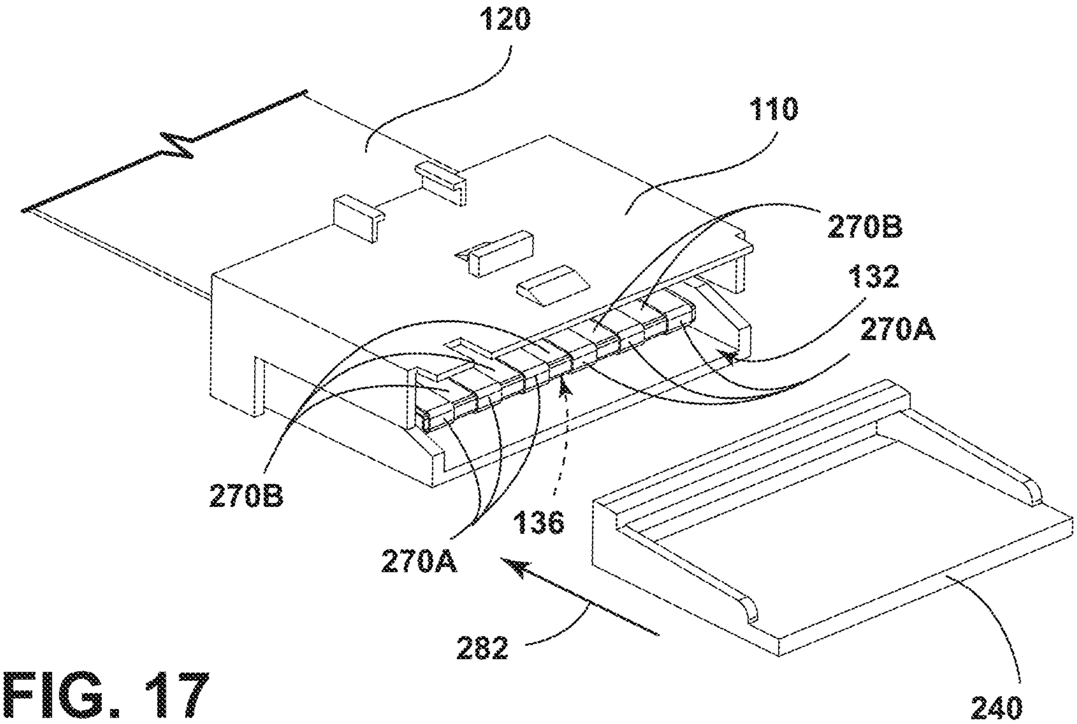
120
110
270B
132
270A
270B
136
270A
282
240

FIG. 17

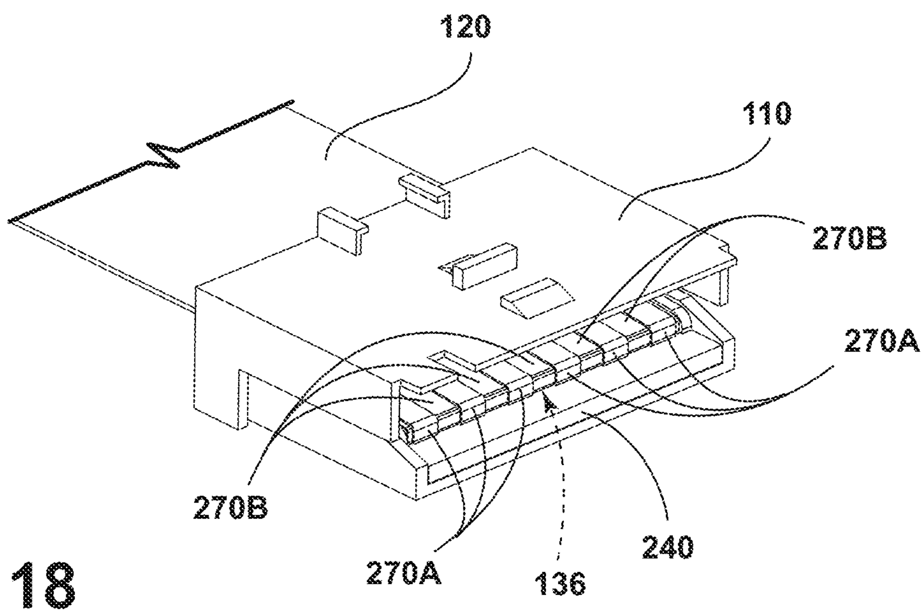
120
110
270B
270A
270B
270A
136
240

FIG. 18

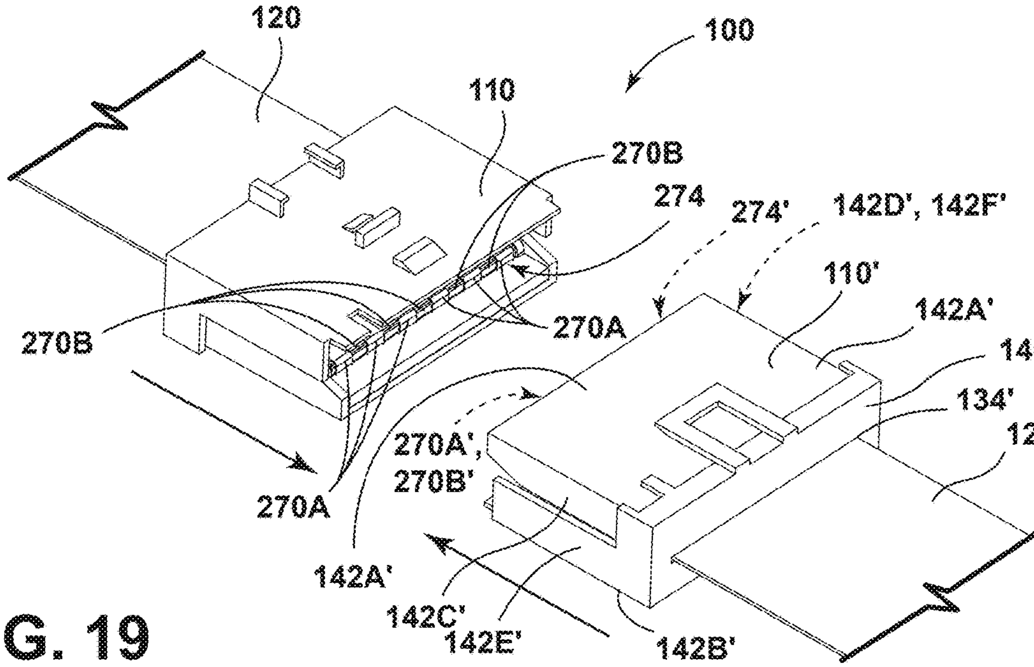

FIG. 19

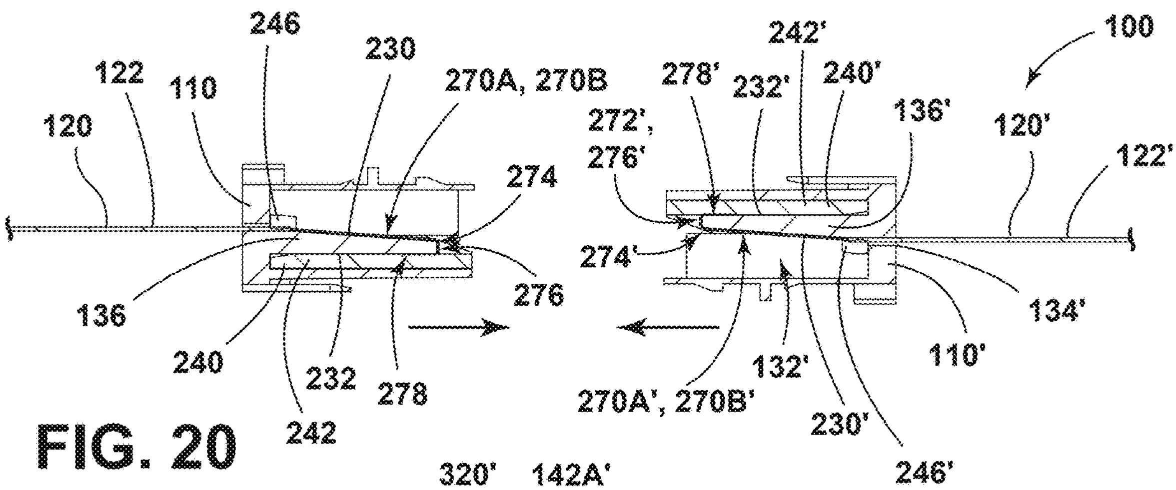

FIG. 20

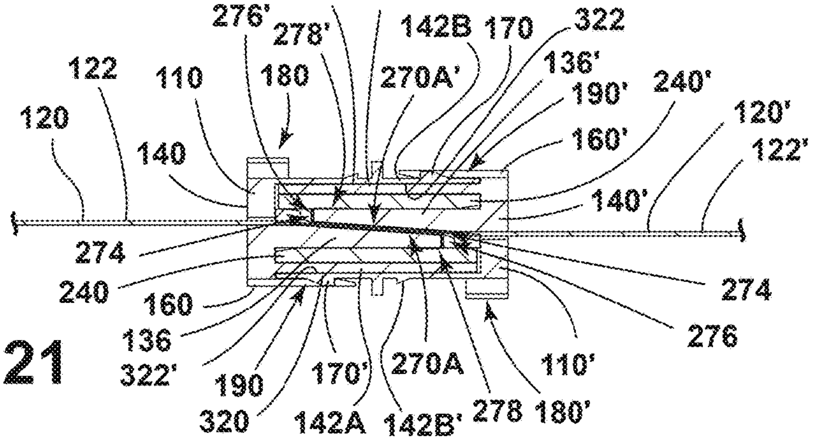

FIG. 21

ELECTRICAL CONNECTOR FOR CONNECTING TO A FLAT CABLE AND ELECTRICAL ASSEMBLY USING SAME

TECHNICAL FIELD

The present disclosure generally relates to electrical assemblies including electrical connectors and/or flat cables that may, for example, be utilized in connection with and/or incorporated into vehicles.

BRIEF DESCRIPTION OF THE DRAWINGS

While the claims are not limited to a specific illustration, an appreciation of various aspects may be gained through a discussion of various examples. The drawings are not necessarily to scale, and certain features may be exaggerated or hidden to better illustrate and explain an innovative aspect of an example. Further, the exemplary illustrations described herein are not exhaustive or otherwise limiting, and embodiments are not restricted to the precise form and configuration shown in the drawings or disclosed in the following detailed description. Exemplary illustrations are described in detail by referring to the drawings as follows:

FIG. **1** is a perspective view generally illustrating an embodiment of an electrical assembly according to teachings of the present disclosure.

FIG. **2** is a side view generally illustrating an embodiment of an electrical assembly according to teachings of the present disclosure.

FIGS. **3** and **4** are perspective views generally illustrating an embodiment of a housing of an electrical connector according to teachings of the present disclosure.

FIG. **5** is a cross-sectional view generally illustrating an embodiment of a housing of an electrical connector according to teachings of the present disclosure.

FIG. **6** is a front view generally illustrating an embodiment of a housing of an electrical connector according to teachings of the present disclosure.

FIGS. **7** and **8** are perspective views generally illustrating an embodiment of a wedge of an electrical connector according to teachings of the present disclosure.

FIG. **9** is a cross-sectional view generally illustrating an embodiment of an electrical connector in an assembled configuration with a flat cable hidden according to teachings of the present disclosure.

FIGS. **10** and **11** are perspective views generally illustrating an embodiment of a method of assembling an electrical assembly according to teachings of the present disclosure.

FIG. **12** is a perspective view generally illustrating an embodiment of a folding tool used in connection with a method of assembling an electrical assembly according to teachings of the present disclosure.

FIGS. **13-19** are perspective views generally illustrating an embodiment of a method of assembling an electrical assembly according to teachings of the present disclosure.

FIGS. **20** and **21** are cross-sectional views generally illustrating an embodiment of a method of assembling an electrical assembly according to teachings of the present disclosure.

DETAILED DESCRIPTION

Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous

specific details are set forth in order to provide a thorough understanding of the various described embodiments. However, it will be apparent to one of ordinary skill in the art that the various described embodiments may be practiced without these specific details. In other instances, well-known methods, procedures, components, circuits, and networks have not been described in detail so as not to unnecessarily obscure aspects of the embodiments.

FIGS. **1** and **2** present an electrical assembly **100** including an electrical connector **110**, a second electrical connector **110'**, a flat cable **120**, and/or a second flat cable **120'**. The electrical connector **110** is configured for connection with the flat cable **120** and/or the second electrical connector **110'**. The second electrical connector **110'** is configured for connection with the second flat cable **120'** and/or the electrical connector **110**. In an assembled configuration, the second electrical connector **110'** is connected with the electrical connector **110** such that the flat cable **120** is electrically connected to the second flat cable **120'**. In some embodiments, the second electrical connector **110'** includes an identical configuration as the electrical connector **110**. For example and without limitation, an electrical connector **110** (e.g., a housing **130** and/or a wedge **240**) may be reversible such that the electrical connector **110**, including the housing **130** and the wedge **240**, are configured to mate with a second electrical connector **110'** having the same configuration to electrically connect the cable **120** with the second cable **120'**. In some configurations, the second electrical connector **110'** may be rotated 180 degrees about an X-axis and 180 degrees about a Y-axis relative to the electrical connector **110**, and then mated with the electrical connector **110** such that portions of the electrical connectors **110**, **110'** are inserted into each other and/or such that one or more portions/walls are flush. Embodiments of the electrical connector **110** are described below, but the second electrical connector **110'** may include some or all of the same or similar features.

FIGS. **3-6** illustrate an electrical connector **110** including a housing **130** having an internal cavity **132**, a slot **134**, and/or a contact plate **136**. The slot **134** is in communication with the cavity **132** and is configured for receiving a flat cable **120**. The contact plate **136** is disposed in the cavity **132**. The housing **130** includes a first end **138**A, a second end **138**B spaced apart from the first end **138**A, an end wall **140**, and/or a plurality of sidewalls **142**A-F disposed between the first end **138**A and the second end **138**B, such as a first sidewall **142**A, a second sidewall **142**B, a third sidewall **142**C, a fourth sidewall **142**D, a fifth sidewall **142**E, and/or a sixth sidewall **142**F. The end wall **140** is disposed proximate the first end **138**A and includes the slot **134**. The housing **130** is open proximate the second end **138**B such that the cavity **132** is accessible (e.g., to at least partially receive a second electrical connector **110'**). The first sidewall **142**A extends from the end wall **140** towards the second end **138**B. The second sidewall **142**B extends from the end wall **140** towards the second end **138**B and is spaced apart from the first sidewall **142**A. The first sidewall **142**A and the second sidewall **142**B may be parallel to an X-Y plane. The third sidewall **142**C extends from the first sidewall **142**A towards the second sidewall **142**B and terminates before the second sidewall **142**B. The fourth sidewall **142**D extends from the first sidewall **142**A towards the second sidewall **142**B and terminates before the second sidewall **142**B. The fourth sidewall **142**D is spaced apart from the third sidewall **142**C. The fifth sidewall **142**E extends from the second sidewall **142**B towards the first sidewall **142**A. The fifth sidewall **142**E includes a first portion **144**A and a second portion **144**B connected to the first portion **144**A.

The first portion **144**A terminates before the first sidewall **142**A and the second portion **144**B extends to the first sidewall **142**A. The sixth sidewall **142**F extends from the second sidewall **142**B towards the first sidewall **142**A and is spaced apart from the fifth sidewall **142**E. The sixth sidewall **142**F includes a first portion **146**A and a second portion **146**B that is connected to the first portion **146**A. The first portion **146**A terminates before the first sidewall **142**A and the second portion **146**B extends to the first sidewall **142**A.

With reference to FIG. **6**, the housing **130** includes an outer surface **150** of the third sidewall **142**C and an outer surface **152** of the fourth sidewall **142**D disposed a first distance D**1** from each other (e.g., in a Y-direction). An outer surface **154** of the fifth sidewall **142**E to an outer surface **156** of the sixth sidewall **142**F are disposed a second distance D**2** from each other (e.g., in the Y-direction). In some example configurations, the first distance D**1** is less than the second distance D**2**.

Referring now to FIGS. **3-6**, the housing **130** includes a latch **160**, a protrusion **170**, and/or mount **180**. The latch **160** extends from the end wall **140** and/or is disposed adjacent to and/or spaced apart from the first sidewall **142**A. The latch **160** includes a void **190** (see, e.g., FIG. **1**). The protrusion **170** extends from the second sidewall **142**B (see, e.g., FIG. **3**). The protrusion **170** may include a sloped configuration, among other configurations. The mount **180** extends from the second sidewall **142**B.

With reference to FIGS. **3** and **4**, the mount **180** includes a first member **200**, a second member **202**, a third member **204**, and/or a fourth member **206**. The first member **200** and the second member **202** are disposed proximate the first end **138**A of the housing **130**. The second member **202** is spaced apart from the first member **200** and/or includes a similar configuration as the first member **200**. For example and without limitation, the first and second members **200**, **202** may include opposing L-shaped configurations that provide a channel **208**. The third member **204** and the fourth member **206** are disposed spaced apart from the first end **138**A. The third member **204** may include a sloped configuration, among other configurations. The fourth member **206** is spaced apart from the third member **204** and/or may include a substantially polygonal (e.g., rectangular, etc.) prism configuration. The members **200-206** may extend from the second sidewall **142**B, such as generally in a Z-direction away from the electrical connector **110**.

Referring now to FIG. **2**, the mount **180** is configured for connection with a fastener **210** to connect an electrical connector **110**, **110'** and/or the electrical assembly **100** with an external object **220**. The external object **220** may, for example and without limitation, include a surface, a portion, and/or a component of a vehicle **222**, such as vehicle floor, a vehicle ceiling, and/or a vehicle panel, and/or may include a non-vehicle component, among others.

With reference to FIG. **5**, the contact plate **136** of the housing **130** extends from the end wall **140** towards the second end **138**B of the housing **130**. The contact plate **136** includes a first surface **230** and a second surface **232** spaced apart from the first surface **230**. The first surface **230** slopes downward from the first end **138**A of the housing **130** towards the second end **138**B of the housing **130** such that a thickness T**1** of the contact plate **136** decreases from the first end **138**A towards the second end **138**B. The first surface **230** and the second surface **232** are configured to contact a flat cable **120**.

Referring now to FIGS. **7** and **8**, an electrical connector **110** includes a wedge **240**. A wedge **240** is configured for connection with a contact plate **136**. A wedge **240** is formed separately from the housing **130**. A wedge **240** and the housing **130** are configured to secure a portion of a flat cable **120** in the cavity **132**. A wedge **240** includes a base **242**, a first sidewall **244**A, a second sidewall **244**B, and/or a connecting portion **246**. The base **242** includes a first end **248**A and a second end **248**B spaced apart from the first end **248**A. The first sidewall **244**A and the second sidewall **244**B extend from the base **242**. In some instances, the first sidewall **244**A and the second sidewall **244**B extend from the first end **248**A towards the second end **248**B and terminate before the second end **248**B. The connecting portion **246** is connected to first sidewall **244**A and the second sidewall **244**B. The connecting portion **246** is disposed proximate the first end **248**A and/or is spaced apart from the base **242**. The connecting portion **246** may include a beam or bar configuration, in some examples. The base **242**, the first sidewall **244**A, the second sidewall **244**B, and the connecting portion **246** collectively define an opening **250**. The opening **250** is configured to receive portions of the contact plate **136** and a flat cable **120**. In some examples, a wedge **240** is configured for connection with the contact plate **136** such that, in an assembled configuration, a part of a flat cable **120** is disposed between the base **242** and the second surface **232** of the contact plate **136**, and an additional part of the flat cable **120** is disposed between the connecting portion **246** and the first surface **230** of the contact plate **136** (see, e.g., FIG. **20**).

With reference to FIG. **9**, the internal cavity **132** of the housing **130** includes a first cavity portion **260**A and a second cavity portion **260**B. The first cavity portion **260**A is disposed between a first lateral side surface **262**A of the contact plate **136** and a first interior surface **264**A of the housing **130**. The second cavity portion **260**B is disposed between a second side surface **262**B of the contact plate **136** and a second interior surface **264**B of the housing **130**. In an assembled configuration, a portion of the first sidewall **244**A of a wedge **240** is disposed in the first cavity portion **260**A, and a portion of the second sidewall **244**B of the wedge **240** is disposed within the second cavity portion **260**B.

Referring now to FIG. **10**, a flat cable **120** includes an elongated body **122** having one or more electrical conductors **270**A and/or an insulating material **270**B that may electrically insulate the conductors **270**A from each other and/or other components. The one or more conductors **270**A may be disposed in parallel with each other and/or may be substantially aligned with a common plane. The one or more conductors **270**A may, in some configurations, be integrally formed with the insulating material **270**B. The one or more conductors **270**A are exposed at an end **272** of a flat cable **120**. A second flat cable **120'** may include the same or a similar configuration as the flat cable **120**. For example, the second flat cable **120'** may include one or more conductors **270**A' and/or insulating material **270**B'.

FIGS. **10-21** present a method of assembling an electrical assembly **100**. Referring now to FIG. **10**, the method includes inserting a flat cable **120** into a slot **134** of an electrical connector **110** and/or inserting a second flat cable **120'** into a slot **134** of a second electrical connector **110'**. For instance, an end **272** of the flat cable **120** including one or more exposed conductors **270**A is inserted into the slot **134** of the electrical connector **110**, and an end **272'** of the second flat cable **120'** including one or more exposed conductors **270**A' is inserted into the slot **134** of the second electrical connector **110'**.

With reference to FIG. **11**, the method includes moving the flat cable **120** through the electrical connector **110** (e.g., in a first direction **280**) such that the end **272** of the flat cable

120 is disposed outside of the electrical connector **110**, and/or moving the second flat cable **120'** through the second electrical connector **110'** (e.g., in a first direction **280**) such that the end **272'** of the second flat cable **120'** is disposed outside of the second electrical connector **110'**.

Referring now to FIG. **12**, a folding tool **300** used in connection with some embodiments of the method is shown. The folding tool **300** includes a monolithic body **302** having a first end **304**A and a second end **304**B spaced apart from the first end **304**A. The body **302** includes a first portion **306**, a second portion **308**, a third portion **310**, and/or a slot **312**. The first portion **306**, the second portion **308**, and/or the third portion **310** extend from the first end **304**A towards the second end **304**B. In some instances, the first portion **306** includes a length **306**L (e.g., in an X-direction) that is greater than a length **308**L of the second portion **308**. The slot **312** extends partially into the body **302** and is disposed between the first portion **306** and the second portion **308**. The slot **312** may include a length **312**L (e.g., in a Y-direction) that is substantially equal to or greater than a width **120**W of a flat cable **120**, **120'** (see, e.g., FIG. **14**). The slot **312** is configured to receive a portion of a flat cable **120**, **120'**.

With reference to FIGS. **13** and **14**, the method includes folding, via the folding tool **300**, the end **272** of the flat cable **120**. In some examples, the method includes inserting a portion of an end **272** of a flat cable **120** into the slot **312** of the folding tool **300**. Referring now to FIG. **13**, the folding tool **300** and a flat cable **120** are shown in a first position. In the first position, the second end **304**B of the folding tool **300** is disposed facing the electrical connector **110**. The method may include rotating the folding tool **300** approximately 180 degrees away from the electrical connector **110** to a second position. In some examples, a user may manipulate the third portion **310** of the folding tool **300** to rotate the folding tool **300** from the first position to the second position. Referring now to FIG. **14**, the folding tool **300** and a flat cable **120** are shown in the second position. In the second position, the first end **304**A of the folding tool **300** is disposed facing the electrical connector **110**, **110'** and the flat cable **120**, **120'** includes a folded end **274**. A second cable **120'** may be inserted into a connector **110'** and/or folded via a folding tool **300** in the same or a similar manner as the cable **120**.

With reference to FIGS. **15** and **16**, a folded end **274** of a flat cable **120**, **120'** includes a first portion **276**, **276'** and a second portion **278**, **278'**. The first portion **276**, **276'** may extend substantially orthogonal from a body **122**, **122'** of a flat cable **120**, **120'** and the second portion **278**, **278'** may extend substantially parallel to the body **122**. For example, an end **274**, **274'** of a flat cable **120**, **120'**, when folded, may include a U-shaped configuration that may open toward the electrical connector **110**, **110'** (e.g., in direction **282**, which may be parallel with an X-direction). The method includes disposing the folded end **274** of the flat cable **120** in contact with a contact plate **136** of the electrical connector **110** and/or disposing the folded end **274'** of the second flat cable **120'** in contact with a contact plate **136** of the second electrical connector **110'**. In some instances, a flat cable **120**, **120'** may be moved in a second direction **282** that is opposite the first direction **280** such that a folded end **274**, **274'** of the flat cable **120**, **120'** is in contact with a contact plate **136** of an electrical connector **110**, **110'** (see, e.g., FIG. **15**). The movement in the second direction **282** may include pulling a portion of the flat cable **120**, **120'** back through the slot **134**, **134'** such that the folded end **274**, **274'** is hooked on the contact plate **136**, **136'**. With reference to FIG. **20**, in an assembled configuration, a flat cable **120**, **120'** may be in contact with a first surface **230** and/or a second surface **232** of a contact plate **136** of an electrical connector **110**, **110'**. In some example configurations, a portion of a body **122** of a flat cable **120**, **120'** may be in contact with the first surface **230** and a second portion **278** of a folded end **274** of a flat cable **120**, **120'** may be in contact with the second surface **232**.

Referring now to FIGS. **17** and **18**, the method includes securing the flat cable **120** to the contact plate **136** of the electrical connector **110** via a wedge **240** and securing the second flat cable **120'** to the contact plate **136** of the second electrical connector **110'** via a second wedge **240'**. In some instances, a wedge **240**, **240'** may be inserted into a cavity **132** of an electrical connector **110**, **110'** and/or positioned within the cavity **132** such that a portion of a flat cable **120**, **120'** (e.g., a portion of a body **122**) is disposed between a connecting portion **246** of a wedge **240**, **240'** and a first surface **230** of a contact plate **136** and a portion of a flat cable **120**, **120'** (e.g., a second portion **278** of a folded end **274**) is disposed between a base **242**, **242'** of a wedge **240**, **240'** and surface **232**, **232'** of the contact plate **136** (see, e.g., FIG. **20**).

With reference to FIGS. **19-21**, the method includes connecting the electrical connector **110** to a second electrical connector **110'**. The second electrical connector **110'** may be formed and/or assembled in the same or a similar manner as the electrical connector **110**. For example, a second electrical connector **110'** may include a housing **130'**, an internal cavity **132'**, a slot **134'**, a contact plate **136'**, an end wall **140'**, sidewalls **142**A'-**142**F', a latch **160'**, a protrusion **170'**, a mount **180'**, a void **190'**, and/or a wedge **240'**. The method may include inserting a second flat cable **120'** into the slot **134'** of the second electrical connector **110'** and/or bending an end **274'** of the second flat cable **120'** to provide a first portion **276'** and a second portion **278'** that may have a U-shaped configuration. The electrical connector **110** is connected to the second electrical connector **110'** such that at least one exposed conductor **270**A of the flat cable **120** secured to the contact plate **136** of the electrical connector **110** is electrically connected with at least one exposed conductor **270**A' of the second flat cable **120'** secured to the contact plate **136'** of the second electrical connector **110'** (see, e.g., FIG. **21**).

Referring to FIG. **21**, in an assembled configuration, the contact plate **136** of the electrical connector **110** is disposed adjacent to the contact plate **136'** of the second electrical connector **110'** such that at least one exposed conductor **270**A of the flat cable **120** is in direct contact with and/or electrically connected to at least one exposed conductor **270**A' of the second flat cable **120'**. In some examples, an outer surface **320** of the first sidewall **142**A of the electrical connector **110** is in contact with an inner surface **322'** of the second sidewall **142**B' of the second electrical connector **110'**, and/or an outer surface **320'** of the first sidewall **142**A' of the second electrical connector **110'** is in contact with an inner surface **322** of the second sidewall **142**B' of the electrical connector **110**.

In an assembled configuration, the protrusion **170** of the electrical connector **110** is at least partially disposed within the void **190'** of the latch **160'** of the second electrical connector **110'**, and/or the protrusion **170'** of the second electrical connector **110'** is at least partially disposed within the void **190** of the latch **160** of the electrical connector **110** (see, e.g., FIG. **21**). In some examples, the outer surface **154** of the fifth sidewall **142**E of the electrical connector **110** is aligned flush with the outer surface **154'** of the fifth sidewall **142**E' of the second electrical connector **110'**, and/or the outer surface **156** of the sixth sidewall **142**F of the electrical connector **110** is aligned flush with the outer surface **156'** of the sixth sidewall **142**F' of the second electrical connector **110'** (see, e.g., FIG. **1**). Additionally or alternatively, the third sidewalls **142**C, **142**C' may be aligned with each other (e.g., in a Z-direction), and/or the fourth sidewalls **142**D, **142**D' may be aligned with each other (e.g., in the Z-direction). The third sidewalls **142**C, **142**C' may be disposed inside of and/or in contact with inner surfaces of the fifth sidewalls **142**E', **142**E, respectively. The fourth sidewalls **142**D, **142**D' may be disposed inside of and/or in contact with inner surfaces of the sixth sidewalls **142**F', **142**F, respectively.

Optionally, in an assembled configuration, the electrical connectors **110**, **110'** cooperate to form a generally rectangular prism. In certain examples, the housings **130**, **130'**, when connected, may form six planar outer surfaces and/or only the latches **160**, **160'**, the protrusions **170**, **170'**, and the mounts **180**, **180'** may extend from the outer surfaces of the housings **130**, **130'**.

This disclosure includes, without limitation, the following embodiments:

1. An electrical connector for a flat cable, the electrical connector comprising: a housing, including: an internal cavity; a slot in communication with the internal cavity, the slot receiving said flat cable; and a contact plate disposed in the internal cavity; and a wedge that connects with the contact plate, the wedge formed separately from the housing; wherein the housing and the wedge secure a portion of said flat cable in the cavity.
2. The electrical connector according to embodiment 1, wherein the housing includes: an end wall; a first sidewall extending from the end wall; a second sidewall spaced from the first sidewall; a third sidewall extending from the first sidewall towards the second sidewall; a fourth sidewall spaced apart from the third sidewall; a fifth sidewall extending from the second sidewall towards the first sidewall, the fifth sidewall having a first portion and a second portion, and the first portion terminating before the first sidewall; and a sixth sidewall spaced apart from the fifth sidewall.
3. The electrical connector according to embodiment 1 or 2, wherein a first distance from an outer surface of the third sidewall to an outer surface of the fourth sidewall is less than a second distance from an outer surface of the fifth sidewall to an outer surface of the sixth sidewall; and the third sidewall, the fourth sidewall, the fifth sidewall, and the sixth sidewall are parallel.
4. The electrical connector according to any preceding embodiment, wherein the housing includes: a latch extending from an end wall of the housing and spaced apart from a first sidewall of the housing; and a protrusion extending from a second sidewall of the housing; and wherein in an assembled configuration: the latch engages a second protrusion of a second electrical connector, and the protrusion engages a second latch of said second electrical connector.
5. The electrical connector according to any preceding embodiment, wherein the housing includes a mount that connects with a fastener to connect the electrical connector with an external object.
6. The electrical connector according to any preceding embodiment, wherein: the contact plate extends from a first end of the housing towards a second end of the housing; the contact plate includes a first surface that slopes downward from the first end towards the second end, and a second surface that is spaced from the first surface; and in an assembled configuration, the first surface and the second surface contact said flat cable.
7. The electrical connector according to any preceding embodiment, wherein the wedge includes: a base having a first end and a second end; a first sidewall and a second sidewall extending from the first end towards the second end; and a connecting portion connected to the first sidewall and the second sidewall and spaced apart from the base; wherein in the assembled configuration, the wedge is connected with the contact plate such that a first part of said flat cable is disposed between the base and the second surface of the contact plate and a second part of said flat cable is disposed between the connecting portion and the first surface of the contact plate.
8. The electrical connector according to any preceding embodiment, wherein the housing is reversible such that it can be mated with a second housing of a second connector to connect said flat cable with a second flat cable.
9. The electrical connector according to any preceding embodiment, wherein the internal cavity includes a first cavity portion disposed between a first lateral side surface of the contact plate and a first interior surface of the housing; and the internal cavity includes a second cavity portion disposed between a second lateral side surface of the contact plate and a second interior surface of the housing.
10. The electrical connector according to any preceding embodiment, wherein in an assembled configuration, a portion of a first sidewall of the wedge is disposed in the first cavity portion and a portion of a second sidewall of the wedge is disposed within the second cavity portion.
11. An electrical assembly, comprising: the electrical connector according to any preceding embodiment; a second electrical connector identical to the electrical connector; the flat cable; and a second flat cable; wherein the second electrical connector connects with the second flat cable and the electrical connector; and in an assembled configuration, the second electrical connector is connected to the electrical connector such that the flat cable is electrically connected to the second flat cable.
12. The electrical assembly according to embodiment 11, wherein the flat cable and the second flat cable each include at least one exposed electrical conductor; and in the assembled configuration, (i) the at least one conductor of the flat cable is in contact with the contact plate of the electrical connector, (ii) the at least one conductor of the second flat cable is in contact with the contact plate of the second electrical connector, and (iii) the contact plate of the electrical connector is disposed adjacent to the contact plate of the second electrical connector such that the at least one conductor of the flat cable is electrically connected to the at least one conductor of the second flat cable.
13. The electrical assembly according to embodiments 11 or 12, wherein the electrical connector and the second electrical connector each include a first sidewall and a second sidewall spaced apart from the first sidewall; and in the assembled configuration, (i) an outer surface of the first sidewall of the electrical connector is in contact with an inner surface of the second sidewall of the second electrical connector, and (ii) an outer surface of the first sidewall of the second electrical connector

is in contact with an inner surface of the second sidewall of the electrical connector.

14. The electrical assembly according to any of embodiments 11-13, wherein the electrical connector and the second electrical connector each include a protrusion and a latch having a void; and in the assembled configuration, (i) the protrusion of the electrical connector is at least partially disposed within the void of the latch of the second electrical connector, and (ii) the protrusion of the second electrical connector is at least partially disposed within the void of the latch of the electrical connector.
15. The electrical assembly according to any of embodiments 11-14, wherein the electrical connector and the second electrical connector each include a sidewall and an additional sidewall spaced apart from the sidewall; and in the assembled configuration, (i) an outer surface of the sidewall of the electrical connector is flush with an outer surface of the sidewall of the second electrical connector, and (ii) an outer surface of the additional sidewall of the electrical connector is aligned flush with an outer surface of the additional sidewall of the second electrical connector.
16. A method of assembling the electrical assembly according to embodiment 11, the method comprising: inserting the flat cable into the slot of the electrical connector; folding, via a folding tool, an end of the flat cable; disposing the folded end of the flat cable in contact with the contact plate of the electrical connector; and securing the flat cable to the contact plate via the wedge.
17. The method according to embodiment 16, including: inserting the second flat cable into a slot of the second electrical connector; folding, via the folding tool, an end of the second flat cable; disposing the folded end of the second flat cable in contact with a contact plate of the second electrical connector; and securing the second flat cable to the contact plate via a wedge of the second electrical connector.
18. The method according to embodiment 16 or 17, including, prior to folding the end of the flat cable, moving the flat cable in a first direction through the electrical connector such that the end of the flat cable is disposed outside of the electrical connector.
19. The method according to any of embodiments 16-18, wherein disposing the folded end of the flat cable in contact with the contact plate of the electrical connector includes moving the flat cable in a second direction that is opposite the first direction such that the flat cable is in contact with a first surface and a second surface of the contact plate.
20. The method according to any of embodiments 16-19, including connecting the electrical connector to the second electrical connector such that at least one exposed electrical conductor of the flat cable in contact with the contact plate of the electrical connector is electrically connected with at least one exposed electrical conductor of the second flat cable in contact with the contact plate of the second electrical connector.

Various examples/embodiments are described herein for various apparatuses, systems, and/or methods. Numerous specific details are set forth to provide a thorough understanding of the overall structure, function, manufacture, and use of the examples/embodiments as described in the specification and illustrated in the accompanying drawings. It will be understood by those skilled in the art, however, that the examples/embodiments may be practiced without such specific details. In other instances, well-known operations, components, and elements have not been described in detail so as not to obscure the examples/embodiments described in the specification. Those of ordinary skill in the art will understand that the examples/embodiments described and illustrated herein are non-limiting examples, and thus it can be appreciated that the specific structural and functional details disclosed herein may be representative and do not necessarily limit the scope of the embodiments.

Reference throughout the specification to "examples, "in examples," "with examples," "various embodiments," "with embodiments," "in embodiments," or "an embodiment," or the like, means that a particular feature, structure, or characteristic described in connection with the example/embodiment is included in at least one embodiment. Thus, appearances of the phrases "examples, "in examples," "with examples," "in various embodiments," "with embodiments," "in embodiments," or "an embodiment," or the like, in places throughout the specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more examples/embodiments. Thus, the particular features, structures, or characteristics illustrated or described in connection with one embodiment/example may be combined, in whole or in part, with the features, structures, functions, and/or characteristics of one or more other embodiments/examples without limitation given that such combination is not illogical or non-functional. Moreover, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the scope thereof.

It should be understood that references to a single element are not necessarily so limited and may include one or more of such element. Any directional references (e.g., plus, minus, upper, lower, upward, downward, left, right, leftward, rightward, top, bottom, above, below, vertical, horizontal, clockwise, and counterclockwise) are only used for identification purposes to aid the reader's understanding of the present disclosure, and do not create limitations, particularly as to the position, orientation, or use of examples/embodiments.

Joinder references (e.g., attached, coupled, connected, and the like) are to be construed broadly and may include intermediate members between a connection of elements, relative movement between elements, direct connections, indirect connections, fixed connections, movable connections, operative connections, indirect contact, and/or direct contact. As such, joinder references do not necessarily imply that two elements are directly connected/coupled and in fixed relation to each other. Connections of electrical components, if any, may include mechanical connections, electrical connections, wired connections, and/or wireless connections, among others. The use of "e.g." in the specification is to be construed broadly and is used to provide non-limiting examples of embodiments of the disclosure, and the disclosure is not limited to such examples.

While processes, systems, and methods may be described herein in connection with one or more steps in a particular sequence, it should be understood that such methods may be practiced with the steps in a different order, with certain steps performed simultaneously, with additional steps, and/or with certain described steps omitted.

'One or more' includes a function being performed by one element, a function being performed by more than one element, e.g., in a distributed fashion, several functions being performed by one element, several functions being performed by several elements, or any combination of the above.

It will also be understood that, although the terms first, second, etc. are, in some instances, used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first contact could be termed a second contact, and, similarly, a second contact could be termed a first contact, without departing from the scope of the various described embodiments. The first contact and the second contact are both contacts, but they are not the same contact.

The terminology used in the description of the various described embodiments herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used in the description of the various described embodiments and the appended claims, the singular forms "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term "and/or" as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms "includes," "including," "comprises," and/or "comprising," when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

All matter contained in the above description or shown in the accompanying drawings shall be interpreted as illustrative only and not limiting. Changes in detail or structure may be made without departing from the present disclosure.

What is claimed is:

1. An electrical connector for a flat cable, the electrical connector comprising:
- a housing, including:
- an internal cavity;
- a slot in communication with the internal cavity, the slot receiving said flat cable; and
- a contact plate disposed in the internal cavity; and
- a wedge that connects with the contact plate, the wedge formed separately from the housing;
- wherein the housing and the wedge secure a portion of said flat cable in the cavity; and
- wherein the housing includes:
  - an end wall;
  - a first sidewall extending from the end wall;
  - a second sidewall spaced from the first sidewall;
  - a third sidewall extending from the first sidewall towards the second sidewall;
  - a fourth sidewall spaced apart from the third sidewall;
  - a fifth sidewall extending from the second sidewall towards the first sidewall, the fifth sidewall having a first portion and a second portion, and the first portion terminating before the first sidewall; and
  - a sixth sidewall spaced apart from the fifth sidewall.

2. The electrical connector of claim **1**, wherein the housing includes:
- a latch extending from an end wall of the housing and spaced apart from a first sidewall of the housing; and
- a protrusion extending from a second sidewall of the housing; and
- wherein in an assembled configuration:
- the latch engages a second protrusion of a second electrical connector, and
- the protrusion engages a second latch of said second electrical connector.

3. The electrical connector of claim **1**, wherein the housing includes a mount that connects with a fastener to connect the electrical connector with an external object.

4. The electrical connector of claim **1**, wherein the housing is reversible such that it can be mated with a second housing of a second connector to connect said flat cable with a second flat cable.

5. The electrical connector of claim **1**, wherein the internal cavity includes a first cavity portion disposed between a first lateral side surface of the contact plate and a first interior surface of the housing; and
- the internal cavity includes a second cavity portion disposed between a second lateral side surface of the contact plate and a second interior surface of the housing.

6. The electrical connector of claim **5**, wherein in an assembled configuration, a portion of a first sidewall of the wedge is disposed in the first cavity portion and a portion of a second sidewall of the wedge is disposed within the second cavity portion.

7. An electrical assembly, comprising:
- the electrical connector of claim **1**;
- a second electrical connector identical to the electrical connector;
- the flat cable; and
- a second flat cable;
- wherein the second electrical connector connects with the second flat cable and the electrical connector; and
- in an assembled configuration, the second electrical connector is connected to the electrical connector such that the flat cable is electrically connected to the second flat cable.

8. The electrical assembly of claim **7**, wherein the flat cable and the second flat cable each include at least one exposed electrical conductor; and
- in the assembled configuration, (i) the at least one conductor of the flat cable is in contact with the contact plate of the electrical connector, (ii) the at least one conductor of the second flat cable is in contact with the contact plate of the second electrical connector, and (iii) the contact plate of the electrical connector is disposed adjacent to the contact plate of the second electrical connector such that the at least one conductor of the flat cable is electrically connected to the at least one conductor of the second flat cable.

9. The electrical assembly of claim **7**, wherein the electrical connector and the second electrical connector each include a first sidewall and a second sidewall spaced apart from the first sidewall; and
- in the assembled configuration, (i) an outer surface of the first sidewall of the electrical connector is in contact with an inner surface of the second sidewall of the second electrical connector, and (ii) an outer surface of the first sidewall of the second electrical connector is in contact with an inner surface of the second sidewall of the electrical connector.

10. The electrical assembly of claim **7**, wherein the electrical connector and the second electrical connector each include a protrusion and a latch having a void; and
- in the assembled configuration, (i) the protrusion of the electrical connector is at least partially disposed within the void of the latch of the second electrical connector, and (ii) the protrusion of the second electrical connector is at least partially disposed within the void of the latch of the electrical connector.

11. The electrical assembly of claim **7**, wherein the electrical connector and the second electrical connector each include a sidewall and an additional sidewall spaced apart from the sidewall; and

in the assembled configuration, (i) an outer surface of the sidewall of the electrical connector is flush with an outer surface of the sidewall of the second electrical connector, and (ii) an outer surface of the additional sidewall of the electrical connector is aligned flush with an outer surface of the additional sidewall of the second electrical connector.

12. A method of assembling the electrical assembly of claim **7**, the method comprising:

inserting the flat cable into the slot of the electrical connector;

folding, via a folding tool, an end of the flat cable;

disposing the folded end of the flat cable in contact with the contact plate of the electrical connector; and

securing the flat cable to the contact plate via the wedge.

13. The method of claim **12**, including:

inserting the second flat cable into a slot of the second electrical connector;

folding, via the folding tool, an end of the second flat cable;

disposing the folded end of the second flat cable in contact with a contact plate of the second electrical connector; and

securing the second flat cable to the contact plate via a wedge of the second electrical connector.

14. The method of claim **13**, including connecting the electrical connector to the second electrical connector such that at least one exposed electrical conductor of the flat cable in contact with the contact plate of the electrical connector is electrically connected with at least one exposed electrical conductor of the second flat cable in contact with the contact plate of the second electrical connector.

15. The method of claim **12**, including, prior to folding the end of the flat cable, moving the flat cable in a first direction through the electrical connector such that the end of the flat cable is disposed outside of the electrical connector.

16. The method of claim **15**, wherein disposing the folded end of the flat cable in contact with the contact plate of the electrical connector includes moving the flat cable in a second direction that is opposite the first direction such that the flat cable is in contact with a first surface and a second surface of the contact plate.

17. An electrical connector for a flat cable, the electrical connector comprising:

a housing, including:

an internal cavity;

a slot in communication with the internal cavity, the slot receiving said flat cable; and

a contact plate disposed in the internal cavity; and

a wedge that connects with the contact plate, the wedge formed separately from the housing;

wherein the housing and the wedge secure a portion of said flat cable in the cavity; and

wherein:

the contact plate extends from a first end of the housing towards a second end of the housing;

the contact plate includes a first surface that slopes downward from the first end towards the second end, and a second surface that is spaced from the first surface; and

in an assembled configuration, the first surface and the second surface contact said flat cable.

18. The electrical connector of claim **17**, wherein the housing includes:

an end wall;

a first sidewall extending from the end wall;

a second sidewall spaced from the first sidewall;

a third sidewall extending from the first sidewall towards the second sidewall;

a fourth sidewall spaced apart from the third sidewall;

a fifth sidewall extending from the second sidewall towards the first sidewall, the fifth sidewall having a first portion and a second portion, and the first portion terminating before the first sidewall; and

a sixth sidewall spaced apart from the fifth sidewall.

19. The electrical connector of claim **18**, wherein a first distance from an outer surface of the third sidewall to an outer surface of the fourth sidewall is less than a second distance from an outer surface of the fifth sidewall to an outer surface of the sixth sidewall; and

the third sidewall, the fourth sidewall, the fifth sidewall, and the sixth sidewall are parallel.

20. The electrical connector of claim **17**, wherein the wedge includes:

a base having a first end and a second end;

a first sidewall and a second sidewall extending from the first end towards the second end; and

a connecting portion connected to the first sidewall and the second sidewall and spaced apart from the base;

wherein in the assembled configuration, the wedge is connected with the contact plate such that a first part of said flat cable is disposed between the base and the second surface of the contact plate and a second part of said flat cable is disposed between the connecting portion and the first surface of the contact plate.

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